

Free Space Optical Communications and DataComm calculations.
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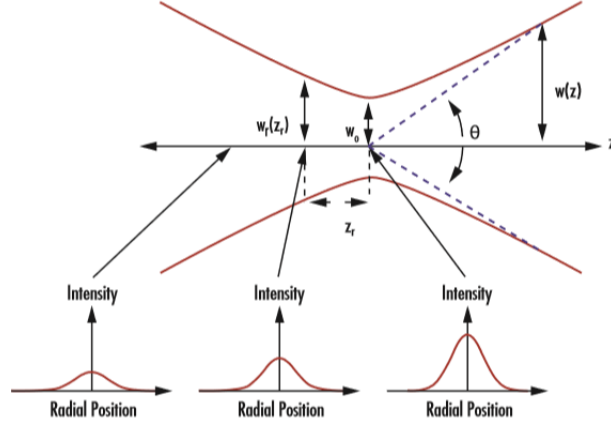
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1 Laser beam propagation

Most of the equations according to Andrew-Phillips book.



In the plot, $W_r(z_r)$ belongs to beam width at Rayleigh distance z_r , W_0 to the beam waist and $W(z)$ to the beam width at any distance z , and θ the beam divergence.

1.1 Divergence

$$\theta = \frac{\lambda}{\pi W_0}$$

1.2 Spot size

With $W_0 = w_0$, the beam spot size is given by:

$$w(z) \approx w_0 + z\theta$$

with differs from beam diameter, which is $2w(z)$ using this convention referred to the graphic.

1.3 Field of View (FOV)

$$\text{FOV} = 2\alpha = 2 \tan^{-1} \left(\frac{d}{2f} \right) \approx \frac{d}{f}$$

1.4 Gaussian beam

$$U_0(r, 0) = \exp \left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0} \right)$$

$$U_0(r, L) = \frac{1}{p(L)} \exp(ikL) \exp \left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0} \right)$$

$$p(L) = 1 - \frac{L}{F_0} + i \frac{2L}{kW_0^2}$$

1.4.1 Gaussian beam parameters

$$\Theta_0 = 1 - \frac{L}{F_0}$$

$$\Lambda_0 = \frac{2L}{kW_0^2}$$

1.4.2 Gaussian Beam width

$$W = W_0 \sqrt{\Theta_0^2 + \Lambda_0^2}$$

1.4.3 Beam width at FWHM and $1/e^2$

$$W_{1/e^2} = \frac{W_{\text{FWHM}}}{\sqrt{2 \ln 2}}$$

1.4.4 Divergence at FWHM and $1/e^2$

$$\theta_{1/e^2} = \frac{2\lambda}{\pi W_0(0)}$$

$$\theta_{\text{FWHM}} = \sqrt{\frac{\ln 2}{2}} \cdot \theta_{1/e^2}$$

1.4.5 Irradiance on-axis

$$I(0, L) = I_0 \cdot \frac{W_0^2}{W^2} \left[\frac{\text{W}}{\text{m}^2} \right]$$

2 Link budget

2.1 Power at the receiver

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{-\frac{2r^2}{w^2}} \right) [\text{W}]$$

2.2 Irradiance at the receiver

$$I_{R_x} = 4 \frac{P_{R_x}}{\pi D_{R_x}^2} \left[\frac{\text{W}}{\text{m}^2} \right]$$

	Component	Symbol	Downlink	Uplink	Unit
Tx	Mean power	P_{Tx}			dB
	Gain	G_{Tx}			dB
	Transmission loss	L_{Tx}			dB
	Pointing loss	L_{PEB}			dB
Channel	Range loss				dB
	Atmospheric attenuation	L_{atm}			dB
	Scintillation loss	L_{sci}			dB
Rx	Gain	G_{Rx}			dB
	Reception loss	L_{Rx}			dB
	Splitting loss				dB
Budget	Power at detector	P_{Rx}			dB
	Required power				dB
	Link margin				dB

3 Atmospheric channel effects

3.1 Kolmogorov outer scale and inner scale

Inertial range:

$$D_{RR}(R) = \langle (V_1 - V_2)^2 \rangle = C_V^2 R^{2/3}$$

$$D_{RR}(R) = \begin{cases} C_v^2 \ell_0^{-4/3} R^2 & , 0 < R \ll \ell_0 \\ C_v^2 R^{2/3} & , \ell_0 \ll R \ll L_0 \end{cases}$$

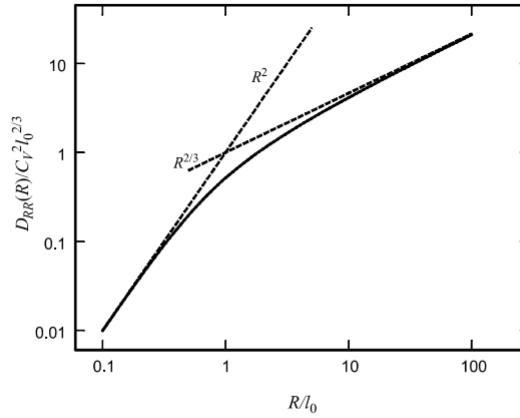


Figure 3.2 The scaled longitudinal velocity structure function showing both R^2 and $R^{2/3}$ asymptotic domains (dashed lines).

3.2 Extended Rytov theory

The received irradiance of an optical wave can be modeled as a modulation process in which small-scale (diffractive) fluctuations are multiplicatively modulated by statistically independent large-scale (refractive) fluctuations:

$$n_{eff}(\mathbf{R}) = n_X(\mathbf{R}) + n_Y(\mathbf{R})$$

X from large scale effects; Y from small scale effects.

$$U(\mathbf{r}, L) = U_0(\mathbf{r}, L) \cdot e^{\psi_X(\mathbf{r}, L)} \cdot e^{\psi_Y(\mathbf{r}, L)}$$

ψ_X, ψ_Y complex phase perturbations on the propagating wave.

- l_o inner scale of turbulence;
- L_o outer scale of turbulence;
- $\sqrt{L/k}$ first Fresnel zone;
- ρ_0 spatial coherence radius of the optical wave;
- $L/k\rho_0$ scattering disk;

Def. scattering disk: refractive cell size l at which the focusing angle $\theta_F \sim l/L$ is equal to the average scattering angle θ_D .

Three regions defined:

- $l_1 \sim \rho_0$; Coherence radius
- $l_2 \sim \sqrt{L/k}$; Fresnel zone
- $l_3 \sim L/k\rho_0$; Scattering disk

Weak fluctuations regime

Fresnel zone < Transverse spatial coherence radius

$$\sqrt{L/k} < \rho_0$$

$$\theta_D \sim \frac{\lambda}{\sqrt{L/k}} \sim 1/\sqrt{L/k}$$

scattering disk on the order of 1st Fresnel zone l_2

Strong fluctuations regime

$$\sqrt{L/k} > \rho_0$$

$$\theta_D \sim \frac{\lambda}{\rho_0} \sim 1/k\rho_0$$

scattering disk $\sim l_3$ when $\sqrt{L/k} \equiv \rho_0$
then $l_1 \sim l_2 \sim l_3$

3.3 Scintillation index model

(...) One random process modulating another: (...) the atmosphere is the modulation process of the instantaneous light that passes through it:

$$\hat{I} = \frac{I}{\langle I \rangle} = XY$$

- X from large scale effects;
- Y from small scale effects;

Second moment

$$\langle \hat{I}^2 \rangle = \langle X^2 \rangle \langle Y^2 \rangle$$

$$= (1 + \sigma_X^2)(1 + \sigma_Y^2)$$

assuming $\langle X \rangle = \langle Y \rangle = 1$ and σ_X^2, σ_Y^2 normalized variances

$$\sigma_I^2 = (1 + \sigma_X^2)(1 + \sigma_Y^2) - 1$$

$$= \sigma_X^2 + \sigma_Y^2 + \sigma_X^2 \sigma_Y^2$$

using Rytov log amplitude of an optical wave (variance of log irradiance), and $\sigma_X^2 = e^{\sigma_{\ln X}^2} - 1$ idem Y , results:

$$\sigma_I^2 = \exp(\sigma_{\ln X}^2 + \sigma_{\ln Y}^2) - 1$$

then in weak irradiance fluctuations results:

$$\sigma_X^2 \cong \sigma_{\ln X}^2$$

$$\sigma_Y^2 \cong \sigma_{\ln Y}^2$$

$$\sigma_I^2 = \sigma_{\ln I}^2 = \sigma_{\ln X}^2 + \sigma_{\ln Y}^2$$

Loss

$$L_{SI} = 4.343 \{ \operatorname{erf}^{-1}(2\rho_{th} - 1) [2 \ln(\sigma_I^2 + 1)]^{\frac{1}{2}} - \frac{1}{2} \ln(\sigma_I^2 + 1) \}$$

3.4 Refractive index structure parameter

3.5 HV: Hufnagel-Valley

$$C_n^2(h) = 0.00594 \left(\frac{w}{27}\right)^2 (10^{-5}h)^{10} \exp\left(-\frac{h}{1000}\right) + 2.7 \times 10^{-16} \exp\left(-\frac{h}{1500}\right) + A \exp\left(-\frac{h}{100}\right)$$

with $A = 1.17 \times 10^{-14} \text{m}^{-2/3}$ and $w = 21 \text{ m/s}$ recommended by ITU for night observations.

3.6 Rytov variance

Also called variance of log irradiance, denotes the strength of scintillation:

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$$

where different turbulence loads are considered as follows:

- $\sigma_R^2 \leq 0.5$ Weak turbulence;
- $0.5 < \sigma_R^2 \leq 5$ Medium to strong;
- $5 < \sigma_R^2 \leq 25$ Strong to saturation regime.

3.7 Fried parameter (Coherence width)

For C_n^2 constant, horizontal propagation with uniform turbulence:

$$r_0 = (0.16 \cdot C_n^2 k^2 L)^{-3/5}$$

For Horizontal link (plane wave):

$$r_0 = (1.46 C_n^2 k^2 L)^{-3/5}$$

For Vertical link or slant paths:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^H C_n^2(h) dh \right]^{-3/5}$$

with ζ the zenith angle ($\sec(0) = 1$), h_0 the height above ground level of the uplink transitter and/or downlink receiver, $H = h_0 + L \cos(\zeta)$ is the satellite altitude.

Downlink can be approximated by a plane wave. Use vertical link formula.

Uplink can be approximated by a spherical wave. Use next formula:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^H C_n^2(h) \left(1 - \frac{h - h_{OGS}}{H - h_{OGS}} \right) dh \right]^{-3/5}$$

with H the satellite altitude, h_{OGS} the altitude of the OGS receiver, and h the integrand.

3.8 Greenwood frequency

$$f_g = 2.31 \lambda^{-6/5} \left[\sec(\theta) \int_{h_0}^H C_n^2(h) V_{\text{wind}}^{5/3}(h) dh \right]^{3/5}$$

4 Link Distance

With R_E the Earth radius, H_{OGS} the ground station height, H or H_{sat} the satellite altitude, and the angles referred to the corresponding figure:

$$L = -(R_E + H_{OGS}) \sin \zeta + \sqrt{(R_E + H)^2 - (R_E + H_{OGS})^2 \cos^2 \zeta}$$

$$L = \sqrt{(R_E + H_{OGS})^2 + (R_E + H_{sat})^2 - 2(R_E + H_{OGS})(R_E + H_{sat}) \cos \gamma}$$

$$d_{max} = -R_E \sin \alpha + \sqrt{(R_E \sin \alpha)^2 + H^2 + 2R_E H}$$

4.1 Beam spreading (Spot size)

Long-term spot size W_{LT} assumed to have long-term Gaussian intensity profile:

$$W_{LT}^2 = W^2 + W^2 T_{SS} + \langle r_c^2 \rangle = W_{ST}^2 + \langle r_c^2 \rangle$$

with r_c the centroid movement, and $\sqrt{\langle r_c^2 \rangle}$ its standard deviation, or beam wander displacement variance.

Loss (Uplink)jki09:

$$L_{SR} = \left[1 + \left(\frac{D_T}{r_0} \right)^{\frac{5}{3}} \right]^{-\frac{6}{5}}$$

4.2 Beam wander

Beam wander variance:

$$\langle r_c^2 \rangle = 0.54 (H - h_{OGS})^2 \sec^2(\zeta) \left(\frac{\lambda}{2W_0} \right)^2 \left(\frac{2W_0}{r_0} \right)^{5/3}$$

assuming an infinite outer scale of turbulence for a collimated Gaussian beam.

$$\langle r_c^2 \rangle = 7.25 L^2 W_0^{-1/3} \csc(\epsilon) \int_{h_0}^H C_n^2(h) \left(1 - \frac{h - h_0}{L \sin \epsilon} \right)^2 dh$$

RMS beam wander displacement:

$$\sqrt{\langle r_c^2 \rangle} = 0.69 \left(\frac{\lambda}{2W_0} \right) \left(\frac{2W_0}{r_0} \right)^{5/6}$$

with r_0 the Fried's parameter.

5 Pointing Error

5.1 Two-line element error

5.2 Star tracker error

5.3 Encoders reading error

5.4 Jitter

5.5 Atmospheric effects

6 Data Communication

6.1 Baud Rate

6.2 Bandwidth

6.3 Modulation

6.4 Spectral Efficiency

6.5 Capacity

$$C = B \cdot \log_2 \left(1 + \frac{S}{N} \right)$$

6.6 Signal-to-Noise Ratio

$$\text{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_S}{2h\nu B}} \rightarrow 10 \cdot \log(\text{SNR}_0) = 74.9 \text{ dB}$$

$$i_S = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\langle i_N^2 \rangle = 2eBi_S = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

6.7 Bit Error Rate

$$\text{4-QAM: } BER \approx \cdot Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

$$\text{16-QAM: } BER \approx 3 \cdot Q \left(\sqrt{\frac{4E_b}{5N_0}} \right)$$

6.8 Mean SNR

$$\langle \text{SNR} \rangle = \frac{\text{SNR}_0}{\sqrt{\left(\frac{P_{s0}}{\langle P_s \rangle}\right) + \sigma_I^2(D_G)\text{SNR}_0^2}}$$

$$\langle \text{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \text{SNR}_0 \rightarrow \infty$$

$$\langle \text{SNR} \rangle = \frac{\langle i_S \rangle}{\sigma_N}$$

6.9 Fading statistics

$$Pr_{\text{fade}} = 1 - \frac{1}{2} \int_0^\infty (u) \text{erfc} \left(\frac{\text{TNR} - \langle \text{SNR} \rangle u}{\sqrt{2}} \right) du$$

$$Pr(E) = \langle \text{BER} \rangle = \frac{1}{2} \int_0^\infty p_f(u) \text{erfc} \left(\frac{\langle \text{SNR} \rangle u}{2\sqrt{2}} \right) du$$

7 Modulation

7.0.1 OOK: On-Off Keying

7.0.2 QPSK: Quadrature Phase Keying

7.0.3 QAM: Quadrature Amplitude Modulation

7.1 Data Rates

8 Photodetectors

8.1 Noise

8.2 APD

8.3 PIN

8.3.1 InGaAS

Example: InGaAs PIN photodiode G6854-01

- Sensitivity $S = 0.95 \text{ A/W}$
- Noise Equivalent Power: $\text{NEP} = 2 \times 10^{-15} \text{ W}/\sqrt{\text{Hz}}$

8.4 QPD

9 Laser transmitter

9.1 Amplitude modulation

9.2 Phase modulation

9.3 Polarization modulation

10 References

?? <https://www.nsnam.org/reviews/2016/socis-midterm/fso.html>